INCH-POUND
MIL-DTL-24643/37E
22 August 2002
SUPERSEDING
MIL-C-24643/37D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, AWG Size No. 18, class B stranding, (7 by 0.0152 inch) uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.
Fourth -	Three conductors cabled together with a nominal lay of 3.5 inches to form each triad.
Fifth -	Identification braid on each triad. Standard identification code applied by method 4.
Sixth -	The specified number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

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Construction (non-watertight) (continued)

Seventh - Binder tape applied helically with overlap.
Eighth - Cross-linked polyolefin jacket. (see table I for thickness).

TABLE I. Details.

Military part no. M24643/37	Size	Number of triads	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LS3U	
				min	max
-01UN	3	3	0.040	0.580	0.620
-02UN	7	7	0.040	0.760	0.810
-03UN	12	12	0.050	1.030	1.090

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor resistance - ohms/1000 feet at 25°C, maximum 7.06
Voltage withstand - volts, root mean square, minimum
Conductor to conductor 2000
Insulator resistance – Megohms-1000 feet minimum
Conductor to conductor 500
Conductor continuity ..... No failure

Group A:
Visual and dimensional..... No failure
Capacitance
Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum 30
Capacitance unbalance - percent maximum 8

Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

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Group C:
Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
- Jacket (cable)
  - Air oven
    - Tensile strength - percent of unaged, minimum 60
    - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged, minimum 50
- Shrinkage ..... No failure
- Heat distortion - percent of unaged, maximum 30
- Permanence of printing (jacket) - cycles, minimum 125

Group D:
- Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80

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Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-035)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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